1. A particle is moving with constant speed in a circular path. When the particle turns by an angle 90°, the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be

[2023 (06 Apr Shift 1)]

(1) 2

(2) 5

(3) 1

(4) 7

2. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: When a body is projected at an angle 45°, its range is maximum.

Reason R: For maximum range, the value of $\sin 2\theta$ should be equal to one.

In the light of the above statements, choose the correct answer from the options given below:

[2023 (06 Apr Shift 1)]

(1) A is false but R is true

(2) A is true but R is false

(3) Both A and R are correct and R is the correct explanation of A

(4) Both A and R are correct but R is NOT the correct explanation of A

3. A child of mass $5\ \mathrm{kg}$ is going round a merry-go-round that makes 1 rotation in $3.14\ \mathrm{s}$. The radius of the merry-go-round is $2\ \mathrm{m}$. The centrifugal force on the child will be

[2023 (06 Apr Shift 2)]

(1) $80\ \mathrm{N}$

(2) $40\ \mathrm{N}$

(3) $100\ \mathrm{N}$

(4) $50\ \mathrm{N}$

4. Two projectiles A and B are thrown with initial velocities of $40\ \mathrm{m\ s^{-1}}$ and $60\ \mathrm{m\ s^{-1}}$ at angles 30° and 60° with the horizontal respectively. The ratio of their ranges respectively is $\left(\mathrm{g} = 10\ \mathrm{m\ s^{-2}}\right)$

[2023 (08 Apr Shift 1)]

(1) $4 : 9$

(2) $2 : \sqrt{3}$

(3) $\sqrt{3} : 2$

(4) $1 : 1$

5. The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$. Where x and y are measured in meter. The maximum height attained by the projectile will be.

[2023 (08 Apr Shift 2)]

(1) $200\ \mathrm{m}$

(2) $10\ \mathrm{m}$

(3) $5\ \mathrm{m}$

(4) $10\sqrt{2}\ \mathrm{m}$

6. The range of the projectile projected at an angle of 15° with horizontal is $50\ \mathrm{m}$. If the projectile is projected with same velocity at an angle of 45° with horizontal, then its range will be

[2023 (10 Apr Shift 1)]

(1) $100\ \mathrm{m}$

(2) $100\sqrt{2}\ \mathrm{m}$

(3) $50\sqrt{2}\ \mathrm{m}$

(4) $50\ \mathrm{m}$

7. Two projectiles are projected at 30° and 60° with the horizontal with the same speed. The ratio of the maximum height attained by the two projectiles respectively is:

[2023 (10 Apr Shift 2)]

(1) $\sqrt{3} : 1$

(2) $1 : \sqrt{3}$

(3) $2 : \sqrt{3}$

(4) $1 : 3$

8. A projectile fired at 30° to the ground is observed to be at same height at time $3\ \mathrm{s}$ and $5\ \mathrm{s}$ after projection, during its flight. The speed of projection of the projectile is $\mathrm{m\ s^{-1}}$.

(Given $g = 10\ \mathrm{m\ s^{-2}}$)

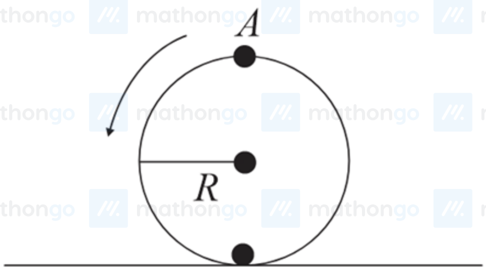
[2023 (11 Apr Shift 1)]

9. A projectile is projected at 30° from horizontal with initial velocity $40 \mathrm{~m} \mathrm{~s}^{-1}$. The velocity of the projectile at $\mathrm{t}=2 \mathrm{~s}$ from the start will be:

[2023 (11 Apr Shift 2)]

(1) $40 \sqrt{3} \mathrm{~m} \mathrm{~s}^{-1}$
(2) Zero
(3) $20 \mathrm{~m} \mathrm{~s}^{-1}$
(4) $20 \sqrt{3} \mathrm{~m} \mathrm{~s}^{-1}$

10. A disc is rolling without slipping on a surface. The radius of the disc is R. At $t=0$, the top most point on the disc is A as shown in figure. When the disc completes half of its rotation, the displacement of point A from its initial position is



[2023 (13 Apr Shift 1)]

(1) $2 R$
(2) $R \sqrt{\left(\pi^2+4\right)}$
(3) $\mathrm{R} \sqrt{\left(\pi^2+1\right)}$
(4) $2 R \sqrt{\left(1+4 \pi^2\right)}$

11. A vehicle of mass $200 \mathrm{~kg}$ is moving along a levelled curved road of radius $70 \mathrm{~m}$ with angular velocity of $0.2 \mathrm{~rad} \mathrm{~s}^{-1}$. The centripetal force acting on the vehicle is:

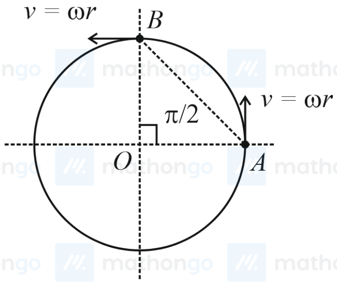
[2023 (13 Apr Shift 2)]

(1) $560 \mathrm{~N}$
(2) $2800 \mathrm{~N}$
(3) $2240 \mathrm{~N}$
(4) $14 \mathrm{~N}$

ANSWER KEYS

1. (1)	**2.** (3)	**3.** (2)	**4.** (1)	**5.** (3)	**6.** (1)	**7.** (4)	**8.** (80)
9. (4)	**10.** (2)	**11.** (1)					

1. (1)



In circular motion with constant speed, magnitude of instantaneous velocity, $v_{ins} = \omega r$.

Time taken to turn by 90° will be $= \frac{\frac{\pi}{2}}{\omega} = \frac{\pi}{2\omega}$.

Displacement will be equal to the length of $AB = \sqrt{r^2 + r^2} = r\sqrt{2}$

Average velocity

$v_{avg} = \frac{\text{displacement}}{\text{time}}$

$= \frac{r\sqrt{2} \times 2\omega}{\pi} = \frac{2\sqrt{2}}{\pi}\omega r$

Hence, required ratio $\frac{v_{\text{ins}}}{v_{\text{avg}}} = \frac{\omega r \pi}{2\sqrt{2}\omega r} = \frac{\pi}{2\sqrt{2}}$

Therefore, $x = 2$.

2. (3)

Horizontal range of a projectile from a horizontal plane is given by, $R = \frac{u^2 \sin 2\theta}{g}$.

Clearly for maximum range, $\sin 2\theta = 1$.

Therefore, $\theta = 45^{\circ}$.

3. (2)

The angular speed (ω) of the child can be calculated as follows:

$\omega = \frac{2\pi}{3.14}$

$= 2 \text{ rad s}^{-1}$

The formula to calculate the centrifugal force on the child is given by

$F_r = m\omega^2 r \quad \ldots(1)$

Substitute the values of the known parameters into equation (1) to calculate the required centrifugal force.

$F_r = 5 \text{ kg} \times \left(2 \text{ rad s}^{-1}\right)^2 \times 2 \text{ m}$

$= 40 \text{ N}$

4. (1)

The range (R_1) of the first projectile is given by

$R_1=\frac{u_1{}^2\sin 2\theta_1}{g} \quad \ldots(1)$

The range (R_2) of the second projectile is given by

$R_2=\frac{u_2{}^2\sin 2\theta_2}{g} \quad \ldots(2)$

Divide equation (1) by equation (2) and simplify to obtain the ratio of the ranges.

$$\frac{R_1}{R_2}=\frac{\frac{u_1{}^2\sin 2\theta_1}{g}}{\frac{u_2{}^2\sin 2\theta_2}{g}}$$
$$=\frac{u_1{}^2\sin 2\theta_1}{u_2{}^2\sin 2\theta_2} \quad \ldots(3)$$

Substitute the values of the known parameters into equation (3) to calculate the required ratio.

$$\frac{R_1}{R_2}=\frac{40^2\times\sin 60^{\circ}}{60^2\times\sin 120^{\circ}}$$
$$=\frac{4}{9}$$

5. (3)

The trajectory of a projectile is parabolic in nature. At maximum height, it can be written that

$\frac{dy}{dx}=0 \quad \ldots(1)$

Substitute the expression for trajectory into equation (1) and solve to calculate the horizontal distance, for which the projectile attains the maximum height.

$$\frac{d}{dx}\left[x-\frac{x^2}{20}\right]=0$$
$$\Rightarrow 1-\frac{x}{10}=0$$
$$\Rightarrow x=10\text{ m}$$

Substitute the value of x in the given expression for the trajectory to obtain the maximum height.

$$y=\left(10-\frac{10^2}{20}\right)\text{ m}$$
$$=5\text{ m}$$

6. (1)

The data given is

$\theta_1=15^{\circ}$

$\theta_2=45^{\circ}$

$R_1=50\text{ m}$

The formula for range of a projectile is given by

$R=\frac{u^2\sin 2\theta}{g} \quad \ldots(i)$

Substituting the values in equation (i)

$$R_1=\frac{u^2\sin 2\theta_1}{g}$$
$$\Rightarrow 50=\frac{u^2\sin 30^{\circ}}{g}$$
$$\Rightarrow \frac{u^2}{g}=100$$

Let the new range be R_2. The magnitude of the range is

$$R_2=\frac{u^2\sin 2\theta_2}{g}$$
$$\Rightarrow R_2=\left(\frac{u^2}{g}\right)\sin 90^{\circ}$$
$$\Rightarrow R_2=100\text{ m}$$

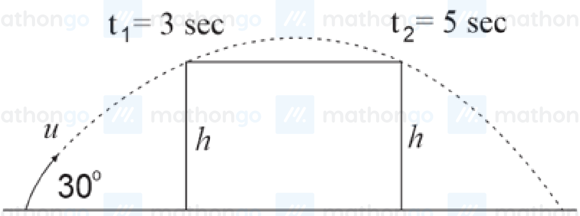
7. (4)

The maximum height is given by, $H=\frac{u^2 \sin^2 \theta}{2g}$.

The given data is
$\theta_1=30^{\circ}$
$\theta_2=60^{\circ}$
$u_1=u_2$
Therefore,

$\frac{(H_1)_{\max}}{(H_2)_{\max}}=\frac{u_1^2 \sin^2 \theta_1}{u_2^2 \sin^2 \theta_2}$
$=\left(\frac{\sin 30^{\circ}}{\sin 60^{\circ}}\right)^2$
$=\left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)^2$
$=\frac{1}{3}$

8. (80)

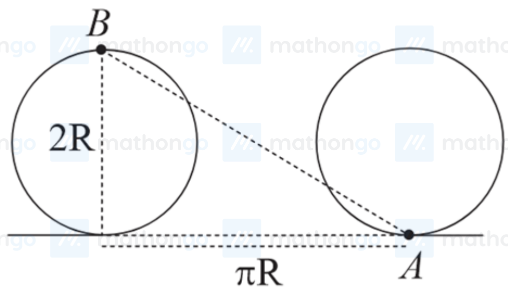


The time taken to reach maximum height is
$T'=\frac{u \sin \theta}{g}$.
The total time of flight is $T=(3+5)\text{ s}=8\text{ s}$.
Hence,
$T'=\left(\frac{T}{2}\right)=\left(\frac{3+5}{2}\right)=\left(\frac{u \sin \theta}{g}\right)$
$\Rightarrow 4=\frac{(u) \times \frac{1}{2}}{10}$
$\Rightarrow u=80\text{ m s}^{-1}$

9. (4)

It is given that $v=40\text{ m s}^{-1}$.
The components of the velocity will be
$v_y=v \sin 30^{\circ}=\frac{40\text{ m s}^{-1}}{2}=20\text{ m s}^{-1}$
$v_x=v \cos 30^{\circ}=\frac{40 \sqrt{3}}{2}\text{ m s}^{-1}=20 \sqrt{3}\text{ m s}^{-1}$
The time taken to reach maximum height is $T=\frac{u \sin 30^{\circ}}{g}=\frac{20\text{ m s}^{-1}}{10\text{ m s}^{-2}}=2\text{ s}$
Hence, at $T=2\text{ s}$, $as\ v_y=0\text{ m s}^{-1}$ therefore $v_{net}=v_x=20 \sqrt{3}\text{ m s}^{-1}$.

10. (2)



The displacement is the hypotenuse of the diameter and the distance travelled for half the rotation πR.
By using Pythagoras theorem,

Displacement of $A=\sqrt{(2R)^2+(\pi R)^2}$

$=R\sqrt{\pi^2+4}$

11. (1)

The centripetal force is given by

$F=\frac{Mv^2}{R}=MR\omega^2$

The given data is

$M=200\ \text{kg}$

$R=70\ \text{m}$

$\omega=0.2\ \text{rad s}^{-1}$

The magnitude of the force is

$F=200\times70\times0.2^2\ \text{N}$

$\Rightarrow F=560\ \text{N}$